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Vector in Java

=> Vector :-

- ➔ Vector is an implementation class of List interface which is present in java.util package
- ➔ Syntax : public class Vector extends AbstractList implements List, RandomAccess, Cloneable, Serializable { - }
- ➔ The underline data structure of Vector is resizable array or growable array
- ➔ Vector was introduced in JDK 1.0 version
- ➔ Vector class is also known as legacy class. (Legacy class is the class which was formed in previous version and was restructured or re-engineered in new version)

-> Properties of Vector :-

1. Vector is an index based Data Structure which means that first element will be inserted at 0 index position
2. Vector can store different data types elements or hetrogeneous elements
3. We can store duplicate elements in the Vector

4. We can store any number of null values in the Vector
5. Vector follows the insertion order which means the sequence in which we are inserting the elements, in the same sequence we can retrieve the elements
6. Vector does not follow the sorting order (above properties are same as List interface)

7. Vector is synchronized collection because Vector contains many synchronized method
8. Vector does not allow more than one thread at one time
9. Vector does not allow parallel execution or Vector allows sequential execution
10. Vector increases the execution time which in turn makes the application slow
11. Vector is threadsafe
12. Vector gurantee for data consistency

-> Working of Vector :-

1. When we create a vector, a vector of 10 initial capacity is created
2. When the vector is full, then new vector will be created automatically with new capacity = current capacity * 2;
3. When new vector is created then all the elements from old vector will copied to new vector and then the reference variable will point to the new vector and garbage collector will delete the previous vector from the memory

Note : In vector we can find the capacity

-> Constructors of Vector :-

1. public Vector()
2. public Vector(int capacity)
3. public Vector(int capacity, int incremental_ratio)
4. public Vector(Collection c)

-> Methods :-

1. Vector contains all the methods of Collection interface

2. Vector contains all the methods of List interface

1. public synchronized int capacity()

2. public synchronized void addElement(Object obj)

3. public synchronized Object firstElement()

4. public synchronized Object lastElement()

5. public synchronized boolean removeElement(Object obj)

6. public synchronized void removeElementAt(int index)

7. public synchronized void removeAllElements()

-> When we should use Vector ?

= We should use Vector in case of retrieval or searching operations

(Vector inherits the RandomAccess interface)

-> When we should not use Vector ?

= We should not use Vector in case of insertion or deletion of elements

-> What is difference between ArrayList & Vector :-

1. ArrayList was introduced in JDK 1.2 version

   Vector was introduced in JDK 1.0 version

2. ArrayList is not a legacy class

   Vector is legacy class

3. ArrayList is non-synchronized collection

Vector is synchronized collection

4. ArrayList allows more than one thread at one time

Vector does not allow more than one thread at one time

5. ArrayList allows the parallel execution

Vector does not allow parallel execution

6. ArrayList decreases the execution time which in turn makes the application fast

Vector increases the execution time which in turn makes the application slow

7. ArrayList is not threadsafe

Vector is threadsafe

8. ArrayList does not gurantee for data consistency

Vector gurantee for data consistency

9. In case of ArrayList new capacity = (present capacity * 3/2)+1

In case of Vector new capacity = present capacity * 2

10. In ArrayList we cannot find the capacity

In Vector we can find the capacity

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